# 20

# Chance Errors in Sampling

*To all the ladies present and some of those absent.*
—THE TOAST USUALLY PROPOSED BY JERZY NEYMAN

## 1. INTRODUCTION

Sample surveys involve chance error. This chapter will explain how to find the likely size of the chance error in a percentage, for simple random samples from a population whose composition is known. That mainly depends on the size of the sample, not the size of the population. First, an example. A health study is based on a representative cross section of 6,672 Americans age 18 to 79. A sociologist now wishes to interview these people. She does not have the resources to do them all, in fact she only has enough money to sample 100 of them. To avoid bias, she is going to draw the sample at random. In the imaginary dialogue which follows, she is discussing the problem with her statistician.[1]

*Soc.* I guess I have to write all the 6,672 names on separate tickets, put them in a box, and draw out 100 tickets at random. It sounds like a lot of work.

*Stat.* We have the files on the computer, code-numbered from 1 to 6,672. So you could just draw 100 numbers at random in that range. Your sample would be the people with those code numbers.

*Soc.* Yes, but then I still have to write the numbers from 1 to 6,672 on the tickets. You haven't saved me much time.

*Stat.* That isn't what I had in mind. With a large box, it's hard to mix the tickets properly. If you don't, most of the draws probably come from the tickets you put in last. That could be a serious bias.

*Soc.* What do you suggest?

*Stat.* The computer has a random number generator. It picks a number at random from 1 to 6,672. The person with that code number goes into the sample. Then it picks a second code number at random, different from the first. That's the second person to go into the sample. The computer keeps going until it gets 100 people. Instead of trying to mix the tickets yourself, let the random numbers do the mixing. Besides, the computer saves all that writing.

*Soc.* OK. But if we use the computer, will my sample be representative?

*Stat.* What do you have in mind?

*Soc.* Well, there were 3,091 men and 3,581 women in the original survey: 46% were men. I want my sample to have 46% men. Besides that, I want them to have the right age distribution. Then there's income and education to think about. Of course, what I really want is a group whose attitudes to health care are typical.

*Stat.* Let's not get into attitudes right now. First things first. I drew a sample to show you. Look at table 1. The first person chosen by the computer was female, so was the second. But the third was male. And so on. Altogether, you got 51 men. That's pretty close.

Table 1. One hundred people were chosen at random and classified by sex. Fifty-one were men (M), and 49 were women (F). In the population, the percentages were 46% and 54%.

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| F | F | M | F | M | M | F | M | M | M | M | F | M | M | M | M | F | M | F | F |
| F | M | M | F | M | F | F | M | F | F | M | M | F | F | F | M | F | M | F | M |
| F | M | F | F | M | M | F | M | M | F | M | F | M | F | M | M | M | F | F | F |
| F | M | M | M | F | M | F | M | M | F | M | M | M | M | F | F | F | M | F | M |
| F | M | F | M | M | M | F | F | F | F | M | M | F | M | M | F | F | F | F | F |

*Soc.* But there should only be 46 men. There must be something wrong with the computer.

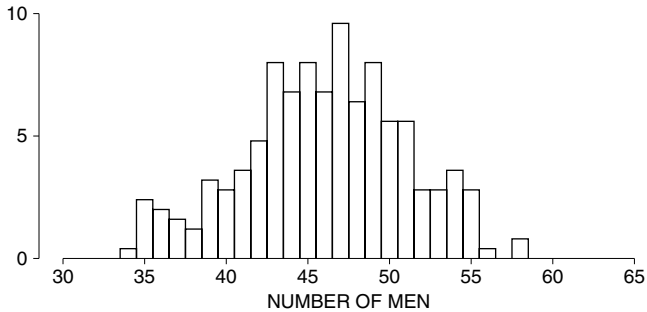
*Stat.* No, not really. Remember, the people in the sample are drawn at random. Just by the luck of the draw, you could get too many men—or too few. I had the computer take a lot of samples for you, 250 in all (table 2). The number of men ranged from a low of 34 to a high of 58. Only 17 samples out of the lot had exactly 46 men. There's a histogram (figure 1).

*Soc.* What stops the numbers from being 46?

*Stat.* Chance variability. Remember the Kerrich experiment I told you about the other day?

*Soc.* Yes, but that was about coin tossing, not sampling.

Figure 1. Histogram for the number of men in samples of size 100.




*Stat.* Well, there isn't much difference between coin tossing and sampling. Each time you toss the coin, you either get a head or a tail, and the number of heads either goes up by one or stays the same. The chances are 50–50 each time. It's the same with sampling. Each time the computer chooses a person for the sample, it either gets a man or a woman, so the number of men either goes up by one or stays the same. The chances are just about 46 to 54 each time—taking 100 tickets out of the box can't change the proportions in the box very much.

*Soc.* What's the point?

*Stat.* The chance variability in sampling is just like the chance variability in coin tossing.

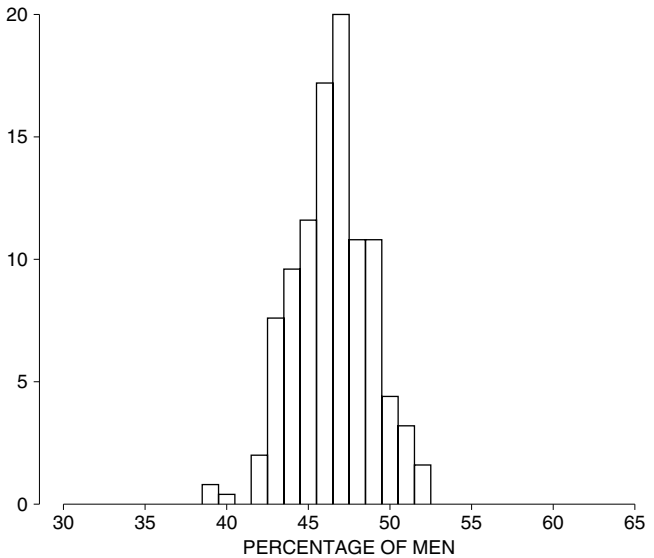
*Soc.* Hmmm. What happens if we increase the size of the sample? Won't it come out more like the population?

*Stat.* Right. For instance, suppose we increase the sample size by a factor of four, to 400. I got the computer to draw another 250 samples, this time with 400 people in each sample. With some of these samples, the percentage of men is below 46%, with others it is above. The low is 39%, the high is 54%.

Table 2. Two hundred fifty random samples were drawn from the respondents to a health study, of whom 46% were men. The sample size was 100. The number of men in each sample is shown below.

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 51 | 40 | 49 | 34 | 36 | 43 | 42 | 45 | 48 | 47 | 51 | 47 | 50 | 54 | 39 | 42 | 47 | 43 | 46 | 46 | 51 | 43 | 53 | 43 | 51 |
| 42 | 49 | 46 | 44 | 55 | 36 | 49 | 44 | 43 | 45 | 42 | 42 | 45 | 43 | 55 | 53 | 49 | 46 | 45 | 42 | 48 | 44 | 43 | 41 | 44 |
| 47 | 54 | 54 | 39 | 39 | 52 | 43 | 36 | 39 | 43 | 43 | 46 | 47 | 44 | 55 | 50 | 53 | 55 | 45 | 43 | 47 | 40 | 47 | 40 | 51 |
| 43 | 56 | 40 | 40 | 49 | 47 | 45 | 49 | 41 | 43 | 45 | 54 | 49 | 50 | 44 | 46 | 48 | 52 | 45 | 47 | 50 | 53 | 46 | 44 | 47 |
| 47 | 46 | 54 | 42 | 44 | 47 | 47 | 36 | 52 | 50 | 51 | 48 | 46 | 45 | 54 | 48 | 46 | 41 | 49 | 37 | 49 | 45 | 50 | 43 | 54 |
| 39 | 55 | 38 | 49 | 44 | 43 | 47 | 51 | 46 | 51 | 49 | 42 | 50 | 48 | 52 | 54 | 47 | 51 | 49 | 44 | 37 | 43 | 41 | 48 | 39 |
| 50 | 41 | 48 | 47 | 50 | 48 | 46 | 37 | 41 | 55 | 43 | 48 | 44 | 40 | 50 | 58 | 47 | 47 | 48 | 45 | 52 | 35 | 45 | 41 | 35 |
| 38 | 44 | 50 | 44 | 35 | 48 | 49 | 35 | 41 | 37 | 46 | 49 | 42 | 53 | 47 | 48 | 36 | 51 | 45 | 43 | 52 | 46 | 49 | 51 | 44 |
| 51 | 51 | 39 | 45 | 44 | 40 | 50 | 50 | 46 | 50 | 49 | 47 | 45 | 49 | 39 | 44 | 48 | 42 | 47 | 38 | 53 | 47 | 48 | 51 | 49 |
| 45 | 42 | 46 | 49 | 45 | 45 | 42 | 45 | 53 | 54 | 47 | 43 | 41 | 49 | 48 | 35 | 55 | 58 | 35 | 47 | 52 | 43 | 45 | 44 | 46 |

Figure 2. Histogram for the percentages of men in samples of size 400. There are 250 samples, drawn at random from the respondents to the health study.




There's a histogram (figure 2). You can compare it with the histogram for samples of size 100. Multiplying the sample size by four cuts the likely size of the chance error in the percentage by a factor of two.

*Soc.* Can you get more specific about this chance error?

*Stat.* Let me write an equation:

$$\text{percentage in sample} = \text{percentage in population} + \text{chance error}.$$

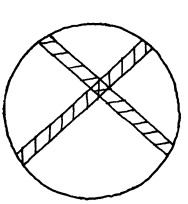
Of course, the chance error will be different from sample to sample—remember the variability in table 2.

*Soc.* So if I let you draw one sample for me, with this random-number business, can you say how big my chance error will be?

*Stat.* Not exactly, but I can tell you its likely size. If you let me make a box model, I can compute the standard error, and then . . . .

*Soc.* Wait. There's one point I missed earlier. How can you have 250 different samples with 100 people each? I mean, $250 \times 100 = 25{,}000$, and we only started with 6,672 people.

*Stat.* The samples are all different, but they have some people in common. Look at the sketch. The inside of the circle is like the 6,672 people, and each shaded strip is like a sample:



The strips are different, but they overlap. Actually, we only scratched the surface with our sampling. The number of different samples of size 100 is over $10^{200}$. That's 1 followed by two hundred 0's. Some physicists don't even think there are that many elementary particles in the whole universe.

## 2. THE EXPECTED VALUE AND STANDARD ERROR

The sociologist of the previous section was thinking about taking a sample of size 100 from a population of 6,672 subjects in a health study. She knew that the percentage of men in the sample would be somewhere around the percentage of men in the population.

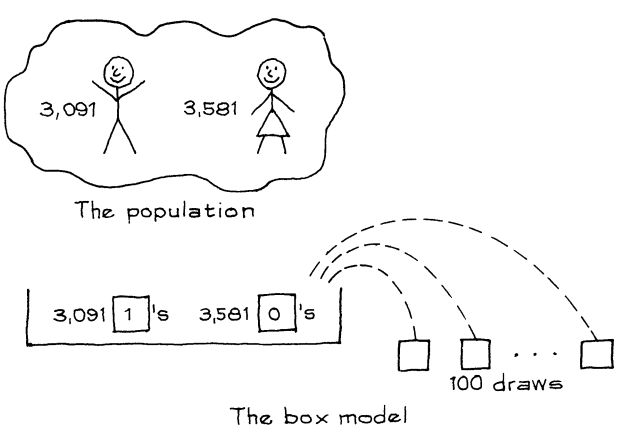
> With a simple random sample, the expected value for the sample percentage equals the population percentage.

However, the sample percentage will not be exactly equal to its expected value—it will be off by a chance error. How big is this error likely to be? The answer is given by the standard error. For the sociologist's problem, the standard error is 5 percentage points. In other words, the sociologist should expect the percentage of men in her sample to be off the percentage in the population by 5 percentage points or so. The method for calculating such standard errors will now be presented. The idea: (i) find the SE for the number of men in the sample; then (ii) convert to percent, relative to the size of the sample. The size of the sample just means the number of sample people—100, in this case.

To compute an SE, you need a box model. The sociologist took a sample of size 100 from a population consisting of 3,091 men and 3,581 women. She classified the people in the sample by sex and counted the men. So there should be only 1's and 0's in the box (section 5 of chapter 17). The number of men in the sample is like the sum of 100 draws from the box

$$\boxed{3{,}091\ \boxed{1}\text{'s}\quad 3{,}581\ \boxed{0}\text{'s}}.$$

She used a simple random sample, so the tickets must be drawn without replacement. This completes the box model.



The fraction of 1's in the box is 0.46. Therefore, the SD of the box is $\sqrt{0.46 \times 0.54} \approx 0.50$. The SE for the sum of 100 draws is $\sqrt{100} \times 0.5 = 5$. The sum of 100 draws from the box will be around 46, give or take 5 or so. In other words, the number of men in the sociologist's sample of 100 is likely to be around 46, give or take 5 or so. The SE for the number of men is 5. Now 46 out of 100 is 46%, and 5 out of 100 is 5%. Therefore, the percentage of men in the sample is likely to be around 46%, give or take 5% or so. This 5% is the SE for the percentage of men in the sample.

> To compute the SE for a percentage, first get the SE for the corresponding number; then convert to percent, relative to the size of the sample. As a cold mathematical formula,
>
> $$\text{SE for percentage} = \frac{\text{SE for number}}{\text{size of sample}} \times 100\%.$$

What happens as the sample gets bigger? For instance, if the sociologist took a sample of size 400, the SE for the number of men in the sample would be

$$\sqrt{400} \times 0.5 = 10.$$

Now 10 represents 2.5% of 400, the size of the sample. The SE for the percentage of men in a sample of 400 would be 2.5%. Multiplying the size of the sample by 4 divided the SE for the percentage by $\sqrt{4} = 2$.

> Multiplying the size of a sample by some factor divides the SE for a percentage not by the whole factor—but by its square root.

The formulas are exact when drawing with replacement. And they are good approximations for draws made without replacement, provided the number of draws is small relative to the number of tickets in the box. For example, take the sociologist's SE. No matter which 100 tickets are drawn, among the tickets left in the box, the percentage of 1's will be very close to 46%. So, as far as the chances are concerned, there isn't much difference between drawing with or without replacement. More about this in section 4.

This section showed how the SE for a percentage can be obtained from the SE for the corresponding number. But these two SEs behave quite differently. When the sample size goes up, the SE for the number goes up—and the SE for the percentage goes down. That is because the SE for the number goes up slowly relative to the sample size (pp. 276, 303):

- The SE for the sample number goes up like the square root of the sample size
- The SE for the sample percentage goes down like the square of the sample size.

## Exercise Set A

1. A town has 30,000 registered voters, of whom 12,000 are Democrats. A survey organization is about to take a simple random sample of 1,000 registered voters. A box model is used to work out the expected value and the SE for the percentage of Democrats in the sample. Match each phrase on list A with a phrase or a number on list B. (Items on list B may be used more than once, or not all.)

| *List A* | *List B* |
|---|---|
| population | number of 1's among the draws |
| population percentage | percentage of 1's among the draws |
| sample | 40% |
| sample size | box |
| sample number | draws |
| sample percentage | 1,000 |
| denominator for sample percentage | 12,000 |

2. A university has 25,000 students, of whom 10,000 are older than 25. The registrar draws a simple random sample of 400 students.
   (a) Find the expected value and SE for the number of students in the sample who are older than 25.
   (b) Find the expected value and SE for the percentage of students in the sample who are older than 25.
   (c) The percentage of students in the sample who are older than 25 will be around _____, give or take _____ or so.

3. A coin will be tossed 10,000 times. Match the SE with the formula. (One formula will be left over.)

| *SE for the . . .* | *Formula* |
|---|---|
| percentage of heads | $\sqrt{10{,}000} \times 50\%$ |
| number of heads | $\dfrac{50}{10{,}000} \times 100\%$ |
| | $\sqrt{10{,}000} \times 0.5$ |

4. Five hundred draws are made at random with replacement from $\boxed{0}\,\boxed{0}\,\boxed{0}\,\boxed{1}$. True or false, and explain:
   (a) The number of 1's among the draws is exactly equal to the sum of the draws.
   (b) The expected value for the percentage of 1's among the draws is exactly equal to 25%.

5. The box $\boxed{0}\,\boxed{0}\,\boxed{0}\,\boxed{1}\,\boxed{2}$ has an average of 0.6, and the SD is 0.8. True or false: the SE for the percentage of 1's in 400 draws can be found as follows—

$$\text{SE for number of 1's} = \sqrt{400} \times 0.8 = 16$$

$$\text{SE for percent of 1's} = \frac{16}{400} \times 100\% = 4\%$$

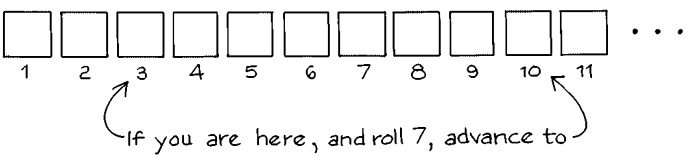
Explain briefly.

6. Nine hundred draws are made at random with replacement from a box which has 1 red marble and 9 blue ones. The SE for the percentage of red marbles in the sample is 1%. A sample percentage which is 1 SE above its expected value equals _____.

$$10\% + 1\% \qquad 1.01 \times 10\%$$

Choose one option, and explain briefly.

7. Someone plays a dice game 100 times. On each play, he rolls a pair of dice, and then advances his token along the line by a number of squares equal to the total number of spots thrown. (See the diagram.) About how far does he move? Give or take how much?




8. According to Sherlock Holmes,

> While the individual man is an insoluble puzzle, in the aggregate he becomes a mathematical certainty. You can, for example, never foretell what any one man will be up to, but you can say with precision what an average number will be up to. Individuals vary, but percentages remain constant. So says the statistician.[2]

The statistician doesn't quite say that. What is Sherlock Holmes forgetting?

*The answers to these exercises are on pp. A79–80.*

*Technical note.* When drawing at random with replacement from a 0–1 box, the SE for the number of 1's among the draws is

$$\sqrt{\text{no. of draws}} \times \text{SD of box}.$$

So the SE for the percentage of 1's among the draws is

$$(\sqrt{\text{no. of draws}} \times \text{SD of box}/\text{no. of draws}) \times 100\%.$$

By algebra, this simplifies to $(\text{SD of box}/\sqrt{\text{no. of draws}}) \times 100\%$. In many books, this would be written $(\sqrt{pq}/\sqrt{n}) \times 100\%$, where $p$ is the fraction of 1's in the box, $q$ is the fraction of 0's, and $n$ is the number of draws.

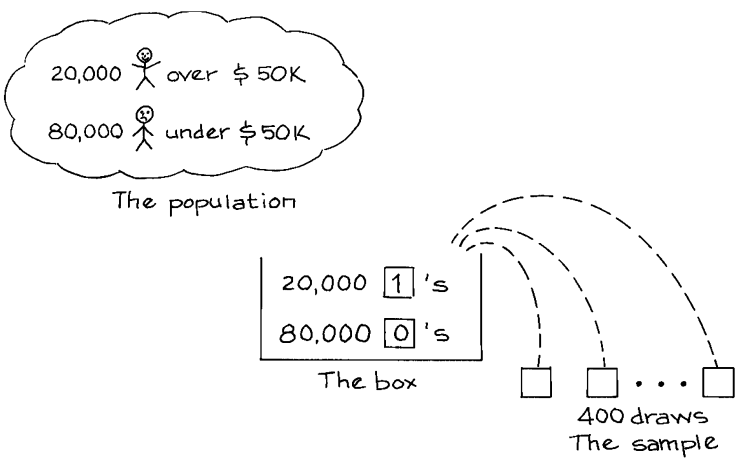
## 3. USING THE NORMAL CURVE

This section will review the expected value and SE for a sample percentage, and use the normal curve to compute chances.

*Example 1.* In a certain town, the telephone company has 100,000 subscribers. It plans to take a simple random sample of 400 of them as part of a market research study. According to Census data, 20% of the company's subscribers earn over $50,000 a year. The percentage of persons in the sample with incomes over $50,000 a year will be around _____, give or take _____ or so.

*Solution.* The first step is to make a box model. Taking a sample of 400 subscribers is like drawing 400 tickets at random from a box of 100,000 tickets. There is one ticket in the box for each person in the population, and one draw for each person in the sample. The drawing is done at random without replacement.

The problem involves classifying the people in the sample according to whether their incomes are more than \$50,000 a year or not, and then counting the ones whose incomes are above that level. So each ticket in the box should be marked 1 or 0. The people earning more than \$50,000 get 1's and the others get 0's. It is given that 20% of the subscribers earn more than \$50,000 a year, so 20,000 of the tickets in the box are marked 1. The other 80,000 are marked 0. The sample is like 400 draws from the box. And the number of people in the sample who earn more than \$50,000 a year is like the sum of the draws. That completes the first step, setting up the box model.




Now you have to work on the sum of the draws from the 0–1 box. The expected value for the sum is $400 \times 0.2 = 80$. To compute the standard error, you need the SD of the box. This is $\sqrt{0.2 \times 0.8} = 0.4$. There are 400 draws, so the SE for the sum is $\sqrt{400} \times 0.4 = 8$. The sum will be around 80, give or take 8 or so. In other words, the number of people in the sample earning more than \$50,000 a year will be around 80, give or take 8 or so.

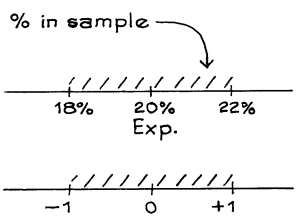
However, the question is about percent. You convert to percent relative to the size of the sample: 80 out of 400 is 20%, and 8 out of 400 is 2%. The expected value for the sample percentage is 20%, and the SE is 2%. That completes the solution: the percentage of high earners in the sample will be around 20%, give or take 2% or so. (It may be unfortunate, but statisticians use the %-sign as an abbreviation both for "percent" and for "percentage point.")

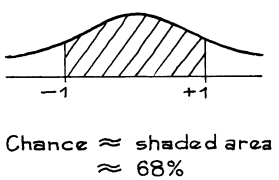
Of course, the expected value for the sample percent is pretty easy to figure, without the detour through the sample number. When drawing at random from a box of 0's and 1's, the expected value for the percentage of 1's among the draws equals the percentage of 1's in the box (p. 359).

> When drawing at random from a box of 0's and 1's, the percentage of 1's among the draws is likely to be around _____, give or take _____ or so. The expected value for the percentage of 1's among the draws fills in the first blank. The SE for the percentage of 1's among the draws fills in the second blank.

*Example 2.* (Continues example 1.) Estimate the chance that between 18% and 22% of the persons in the sample earn more than $50,000 a year.

*Solution.* The expected value for the sample percentage is 20%, and the SE is 2%. Now convert to standard units:




This completes the solution.

Here, the normal curve was used to figure chances. Why is that legitimate? There is a probability histogram for the number of high earners in the sample (figure 3). Areas in this histogram represent chances. For instance, the area between 80 and 90 represents the chance of drawing a sample which has between 80 and 90 high earners. As discussed in chapter 18, this probability histogram follows the normal curve (top panel of figure 3). Conversion to percent is only a change of scale, so the probability histogram for the sample percentage (bottom panel) looks just like the top histogram—and follows the curve too. In example 2, the curve was used on the probability histogram for the sample percentage, not on a histogram for data.

Examples 1 and 2 are about qualitative data. The incomes start out as quantitative data—numbers. However, the problems involve classifying and counting. Each person is classified as earning more than $50,000 a year, or less. Then the high earners are counted. In other words, the data are treated as qualitative: each income either has or doesn't have the quality of being more than $50,000 a year.

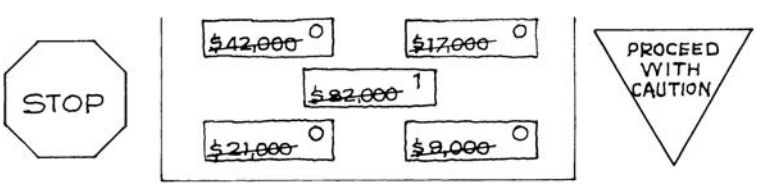
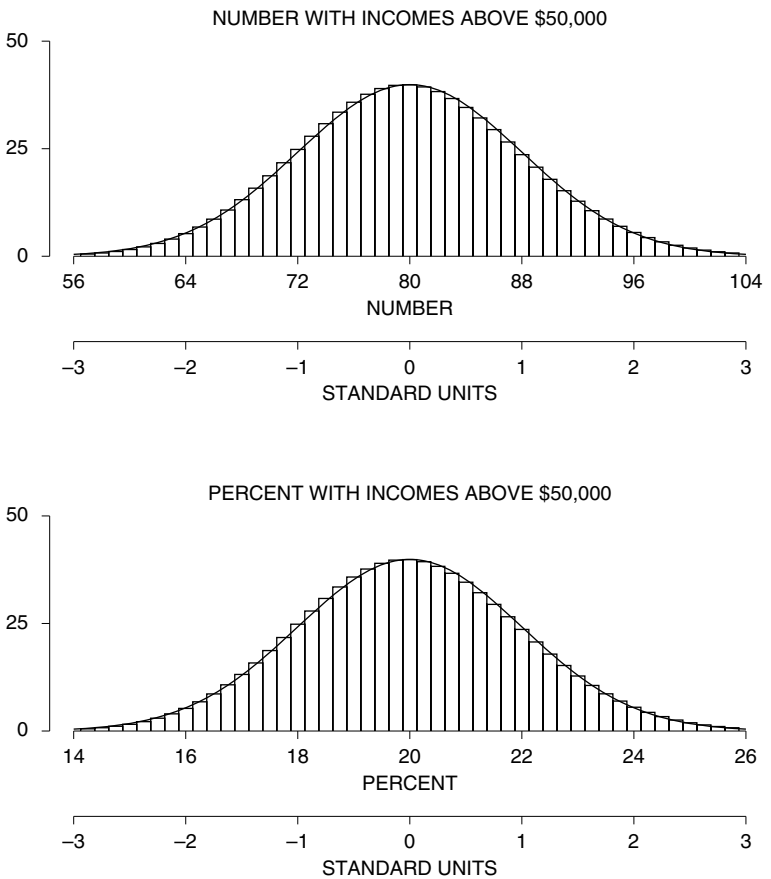


Figure 3. The top panel shows the probability histogram for the number of sample persons with incomes over $50,000. The bottom panel shows the probability histogram for the percentage of sample persons with incomes over $50,000. In standard units, the two histograms are exactly the same.[3] (Four hundred persons are chosen at random from a population of 100,000.)




When do you change to a 0–1 box? To answer this question, think about the arithmetic being done on the sample values. The arithmetic might involve:

- adding up the sample values, to get an average;

or

- classifying and counting, to get a percent.

If the problem is about classifying and counting, put 0's and 1's in the box (section 5 of chapter 17).

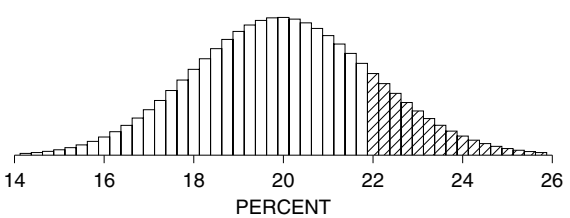
## Exercise Set B

1. You are drawing at random from a large box of red and blue marbles. Fill in the blanks.
   (a) The expected value for the percentage of reds in the _____ equals the percentage of reds in the _____. Options: sample, population
   (b) As the number of draws goes up, the SE for the _____ of reds in the sample goes up but the SE for the _____ of reds goes down. Options: number, percentage

2. In a certain town, there are 30,000 registered voters, of whom 12,000 are Democrats. A survey organization is about to take a simple random sample of 1,000 registered voters.
   (a) The expected value for the percentage of Democrats in the sample is _____. The SE for the percentage of Democrats in the sample is _____.
   (b) The percentage of Democrats in the sample is likely to be around _____, give or take _____ or so.
   (c) Find the chance that between 39% and 41% of the registered voters in the sample are Democrats.

3. According to the Census, a certain town has a population of 100,000 people age 18 and over. Of them, 60% are married, 10% have incomes over $75,000 a year, and 20% have college degrees.[4] As part of a pre-election survey, a simple random sample of 1,600 people will be drawn from this population.
   (a) To find the chance that 58% or less of the people in the sample are married, a box model is needed. Should the number of tickets in the box be 1,600, or 100,000? Explain. Then find the chance.
   (b) To find the chance that 11% or more of the people in the sample have incomes over $75,000 a year, a box model is needed. Should each ticket in the box show the person's income? Explain. Then find the chance.
   (c) Find the chance that between 19% and 21% of the people in the sample have a college degree.

4. The figure below is the probability histogram for the percent of sample persons with incomes above $50,000 (example 1, and bottom panel of figure 3). The shaded area represents _____. Fill in the blank with a phrase.




5. (a) In the top panel of figure 3, the area of the rectangle over 88 represents what?
   (b) In the bottom panel of figure 3, the area of the rectangle over 22% represents what?
   (c) The rectangles in parts (a) and (b) have equal areas. Is that a coincidence?

*The answers to these exercises are on pp. A80–81.*

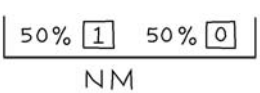
## 4. THE CORRECTION FACTOR

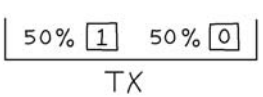
It is just after Labor Day, 2004. The presidential campaign (Bush versus Kerry) is in full swing, and the focus is on the Southwest. Pollsters are trying to predict the results. There are about 1.5 million eligible voters in New Mexico, and about 15 million in the state of Texas. Suppose one polling organization takes a simple random sample of 2,500 voters in New Mexico, in order to estimate the percentage of voters in that state who are Democratic. Another polling organization takes a simple random sample of 2,500 voters from Texas. Both polls use exactly the same techniques. Both estimates are likely to be a bit off, by chance error. For which poll is the chance error likely to be smaller?

The New Mexico poll is sampling one voter out of 600, while the Texas poll is sampling one voter out of 6,000. It does seem that the New Mexico poll should be more accurate than the Texas poll. However, this is one of the places where intuition comes into head-on conflict with statistical theory, and it is intuition which has to give way. In fact, the accuracy expected from the New Mexico poll is just about the same as the accuracy to be expected from the Texas poll.

> When estimating percentages, it is the absolute size of the sample which determines accuracy, not the size relative to the population. This is true if the sample is only a small part of the population, which is the usual case.[5]

A box model will help in focusing the issue. We'll need two boxes, NM and TX. Box NM represents New Mexico, box TX represents Texas. Box NM has 1,500,000 tickets, one for each voter. The tickets corresponding to Democrats are marked 1, the others are marked 0. To keep life simple, we make the percentage of 1's in the box equal to 50%. We hire a polling organization to take a simple random sample from box NM, without telling them what is in the box. (Remember, taking a simple random sample means drawing at random without replacement.) The job of the polling organization is to estimate the percentage of 1's in the box. Naturally, they use the percentage of 1's in their sample.




Now for Box TX. This represents Texas, so it has 15,000,000 tickets. Again, we mark 1 on half the tickets in the box, and 0 on the others. Another polling organization is hired to take a simple random sample of 2,500 tickets from box TX, without knowing the composition of the box. This organization too will estimate the percentage of 1's in the box by the percentage in the sample, and will be off by a chance error.

Box NM and box TX have been set up with the same percentage composition, and the two samples are the same size. Intuition would insist that the organization sampling from box NM will have a much smaller chance error, because

box NM is so much smaller. But statistical theory shows that the likely size of the chance error is just about the same for the two polls.

The issue has now been stated sharply. How does statistical theory justify itself? To begin with, suppose the samples were drawn with replacement. Then it wouldn't matter at all which box was used. There would be a 50–50 chance to get a 0 or a 1 on each draw, and the size of the box would be completely irrelevant. Box NM and box TX have the same SD of 0.5, so both polling organizations would have the same SE for the number of 1's among the draws:

$$\sqrt{2{,}500} \times 0.5 = 25.$$

As a result, they would both have the same SE for the percentage of 1's among the draws:

$$\frac{25}{2{,}500} \times 100\% = 1\%.$$

If they drew at random with replacement, both organizations would be off by about 1 percentage point or so.

In fact, the draws are made without replacement. However, the number of draws is just a tiny fraction of the number of tickets in the box. Taking the draws without replacement barely changes the composition of the box. On each draw, the chance of getting a 1 must still be very close to 50%, and similarly for 0. As far as the chances are concerned, there is almost no difference between drawing with or without replacement.

In essence, that is why the size of the population has almost nothing to do with the accuracy of estimates. Still, there is a shade of difference between drawing with and without replacement. When drawing without replacement, the box does get a bit smaller, reducing the variability slightly. So the SE for drawing without replacement is a little less than the SE for drawing with replacement. There is a mathematical formula that says how much smaller:

$$\begin{array}{c}\text{SE when drawing}\\ \text{WITHOUT replacement}\end{array} = \begin{array}{c}\text{correction}\\ \text{factor}\end{array} \times \begin{array}{c}\text{SE when drawing}\\ \text{WITH replacement}\end{array}$$

The correction factor itself is somewhat complicated:

$$\sqrt{\frac{\text{number of tickets in box} - \text{number of draws}}{\text{number of tickets in box} - \text{one}}}$$

When the number of tickets in the box is large relative to the number of draws,

Table 3. The correction factor; the number of draws is fixed at 2,500.

| *Number of tickets in the box* | *Correction factor (to five decimals)* |
|---:|---:|
| 5,000 | 0.70718 |
| 10,000 | 0.86607 |
| 100,000 | 0.98743 |
| 500,000 | 0.99750 |
| 1,500,000 | 0.99917 |
| 15,000,000 | 0.99992 |

the correction factor is nearly 1 and can be ignored (table 3, p. 368). Then it is the absolute size of the sample which determines accuracy, through the SE for drawing with replacement. The size of the population does not really matter. On the other hand, if the sample is a substantial fraction of the population, the correction factor must be used.

In our box model, the percentage of 1's was the same for both boxes. In reality, the percentage of Democrats will be different for the two states. However, even quite a large difference will generally not matter very much. In the 2004 presidential election, for example, 50% of the voters in New Mexico chose the Republican candidate (Bush), compared to 61% in Texas.[6] But the SDs for the two states are almost the same:

| 50% [1] 50% [0] | 61% [1] 39% [0] |
|---|---|
| NM | TX |
| $SD = \sqrt{.50 \times .50} = .50$ | $SD = \sqrt{.61 \times .39} \approx .49$ |

A sample of size 2,500 will do as well in Texas as in New Mexico, although Texas is 10 times larger. The Texan in the cartoon is just wrong.

"A BIG STATE NEEDS A BIG SAMPLE, PARDNER."

A non-mathematical analogy may help. Suppose you took a drop of liquid from a bottle, for chemical analysis. If the liquid is well mixed, the chemical composition of the drop should reflect the composition of the whole bottle, and it really wouldn't matter if the bottle was a test tube or a gallon jug. The chemist doesn't care whether the drop is 1% or 1/100 of 1% of the solution.

The analogy is precise. There is one ticket in the box for each molecule in the bottle. If the liquid is well mixed, the drop is like a random sample. The number of molecules in the drop corresponds to the number of tickets drawn. This number—the sample size—is so large that chance error in the percentages is negligible.

## Exercise Set C

1. One public opinion poll uses a simple random sample of size 1,500 drawn from a town with a population of 25,000. Another poll uses a simple random sample of size 1,500 from a town with a population of 250,000. The polls are trying to estimate the percentage of voters who favor single-payer health insurance. Other things being equal:

   (i) the first poll is likely to be quite a bit more accurate than the second.
   (ii) the second poll is likely to be quite a bit more accurate than the first.
   (iii) there is not likely to be much difference in accuracy between the two polls.

2. You have hired a polling organization to take a simple random sample from a box of 100,000 tickets, and estimate the percentage of 1's in the box. Unknown to them, the box contains 50% 0's and 50% 1's. How far off should you expect them to be:

   (a) if they draw 2,500 tickets?
   (b) if they draw 25,000 tickets?
   (c) if they draw 100,000 tickets?

3. A survey organization wants to take a simple random sample in order to estimate the percentage of people who have seen a certain television program. To keep the costs down, they want to take as small a sample as possible. But their client will only tolerate chance errors of 1 percentage point or so in the estimate. Should they use a sample of size 100, 2,500, or 10,000? You may assume the population to be very large; past experience suggests the population percentage will be in the range 20%–40%.

4. One hundred draws are made at random with replacement from each of the following boxes. The SE for the percentage of 1's among the draws is smallest for box _____ and largest for box _____. Or is the SE the same for all three boxes?

   (A) $\boxed{\boxed{0}\ \boxed{1}}$ (B) $\boxed{10\ \boxed{0}\text{'s}\ \ 10\ \boxed{1}\text{'s}}$ (C) $\boxed{1{,}000\ \boxed{0}\text{'s}\ \ 1{,}000\ \boxed{1}\text{'s}}$

5. A box contains 2 red marbles and 8 blue ones. Four marbles are drawn at random. Find the SE for the percentage of red marbles drawn, when the draws are made

   (a) with replacement. (b) without replacement.

*The answers to these exercises are on p. A81.*

## 5. THE GALLUP POLL

The Gallup Poll predicts the vote with good accuracy, by sampling several thousand eligible voters out of 200 million. How is this possible? The previous section focused on simple random sampling, but the conclusions hold for most probability methods of drawing samples, including the one used by the Gallup

Poll: the likely size of the chance error in sample percentages depends mainly on the absolute size of the sample, and hardly at all on the size of the population. The huge number of eligible voters makes it hard work to draw the sample, but does not affect the standard error.

Is 2,500 a big enough sample? The square root law provides a benchmark. For example, with 2,500 tosses of a coin, the standard error for the percentage of heads is only 1%. Similarly, with a sample of 2,500 voters, the likely size of the chance error is only a percentage point or so. That is good enough unless the election is very close, like Bush versus Gore in 2000. The Electoral College would be a major complication: the Gallup Poll only predicts the popular vote.

## 6. REVIEW EXERCISES

*Review exercises may also cover material from previous chapters.*

1. Complete the following table for the coin-tossing game.

| | *Number of heads* | | *Percent of heads* | |
|---|---|---|---|---|
| *Number of tosses* | *Expected value* | *SE* | *Expected value* | *SE* |
| 100 | 50 | 5 | 50% | 5% |
| 2,500 | | | | 1% |
| 10,000 | | | | |
| 1,000,000 | | | | |

2. A die is rolled one thousand times. The percentage of aces (⚀) should be around _____, give or take _____ or so.
   (a) The first step in solving this problem is
      (i) computing the SD of the box.
      (ii) computing the average of the box.
      (iii) setting up the box model.
   Choose one option and explain.
   (b) Now solve the problem.

3. A group of 50,000 tax forms has an average gross income of $37,000, with an SD of $20,000. Furthermore, 20% of the forms have a gross income over $50,000. A group of 900 forms is chosen at random for audit. To estimate the chance that between 19% and 21% of the forms chosen for audit have gross incomes over $50,000, a box model is needed.
   (a) Should the number of tickets in the box be 900 or 50,000?
   (b) Each ticket in the box shows

   a zero or a one  a gross income

   (c) True or false: the SD of the box is $20,000.
   (d) True or false: the number of draws is 900.
   (e) Find the chance (approximately) that between 19% and 21% of the forms chosen for audit have gross incomes over $50,000.

(f) With the information given, can you find the chance (approximately) that between 9% and 11% of the forms chosen for audit have gross incomes over $75,000? Either find the chance, or explain why you need more information.

4. As in exercise 3, except it is desired to find the chance (approximately) that the total gross income of the audited forms is over $33,000,000. Work parts (a) through (d); then find the chance or explain why you need more information.

5. (Hypothetical.) On the average, hotel guests who take elevators weigh about 150 pounds with an SD of about 35 pounds. An engineer is designing a large elevator for a convention hotel, to lift 50 such people. If she designs it to lift 4 tons, the chance it will be overloaded by a random group of 50 people is about _____. Explain briefly.

6. The Census Bureau is planning to take a sample amounting to 1/10 of 1% of the population in each state in order to estimate the percentage of the population in that state earning over $100,000 a year. Other things being equal:

   (i) The accuracy to be expected in California (population 35 million) is about the same as the accuracy to be expected in Nevada (population 2 million).
   (ii) The accuracy to be expected in California is quite a bit higher than in Nevada.
   (iii) The accuracy to be expected in California is quite a bit lower than in Nevada.

   Explain.

7. Five hundred draws are made at random from the box

   | 60,000 $\boxed{0}$'s 20,000 $\boxed{1}$'s |

   True or false, and explain:

   (a) The expected value for the percentage of 1's among the draws is exactly 25%.
   (b) The expected value for the percentage of 1's among the draws is around 25%, give or take 2% or so.
   (c) The percentage of 1's among the draws will be around 25%, give or take 2% or so.
   (d) The percentage of 1's among the draws will be exactly 25%.
   (e) The percentage of 1's in the box is exactly 25%.
   (f) The percentage of 1's in the box is around 25%, give or take 2% or so.

8. In a certain town, there are 30,000 registered voters, of whom 12,000 are Democrats. A survey organization is about to take a simple random sample of 1,000 registered voters. There is about a 50–50 chance that the percentage of Democrats in the sample will be bigger than _____. Fill in the blank, and explain.

9. Six hundred draws will be made at random with replacement from the box $\boxed{0}\ \boxed{0}\ \boxed{1}$. The number of 1's among the draws will be around _____ give or take _____ or so.

10. A coin is tossed 2,000 times. Someone wishes to compute the SE for the number of heads among the tosses as $\sqrt{2{,}000} \times 0.5 \approx 22$. Is this the right SE? Answer yes or no, and explain briefly.

11. A university has 25,000 students, of whom 17,000 are undergraduates. The housing office takes a simple random sample of 500 students; 357 out of the 500 are undergraduates. Fill in the blanks.

    (a) For the number of undergraduates in the sample, the observed value is _____ but the expected value is _____.

    (b) For the percentage of undergraduates in the sample, the observed value is _____ but the expected value is _____.

12. There are 50,000 households in a certain city. The average number of persons age 16 and over living in each household is known to be 2.38; the SD is 1.87. A survey organization plans to take a simple random sample of 400 households, and interview all persons age 16 and over living in the sample households. The total number of interviews will be around _____, give or take _____ or so. Explain briefly.

## 7. SUMMARY

1. The sample is only part of the population, so the percentage composition of the sample usually differs by some amount from the percentage composition of the whole population.

2. For probability samples, the likely size of the chance error (the amount off) is given by the standard error.

3. To figure the SE, a box model is needed. When the problem involves classifying and counting, or taking percents, there should only be 0's and 1's in the box. Change the box, if necessary.

4. When drawing at random from a 0–1 box, the expected value for the percentage of 1's in the sample equals the percentage of 1's in the box. To find the SE for the percentage, first get the SE for the corresponding number, then convert to percent. The formula:

$$\text{SE for percentage} = \frac{\text{SE for number}}{\text{size of sample}} \times 100\%.$$

5. When the sample is only a small part of the population, the number of individuals in the population has almost no influence on the accuracy of the sample percentage. It is the absolute size of the sample (that is, the number of individuals in the sample) which matters, not the size relative to the population.

6. The square root law is exact when draws are made with replacement. When the draws are made without replacement, the formula gives a good approximation—provided the number of tickets in the box is large relative to the number of draws.

7. When drawing without replacement, to get the exact SE you have to multiply by the correction factor:

$$\sqrt{\frac{\text{number of tickets in box} - \text{number of draws}}{\text{number of tickets in box} - \text{one}}}$$

When the number of tickets in the box is large relative to the number of draws, the correction factor is nearly one.